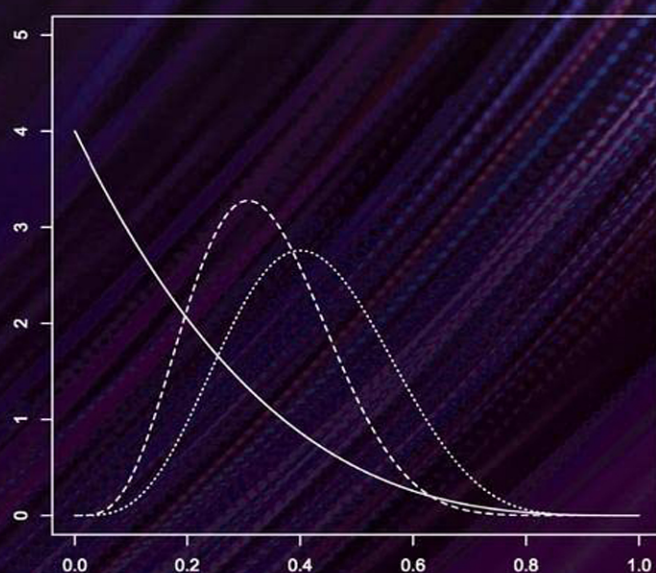


WILEY SERIES IN PROBABILITY AND STATISTICS

THIRD EDITION

PROBABILITY AND STATISTICAL INFERENCE



MAGDALENA NIEWIADOMSKA-BUGAJ
ROBERT BARTOSZYŃSKI



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PROBABILITY AND STATISTICAL INFERENCE

Third Edition

Magdalena Niewiadomska-Bugaj

Robert Bartoszyński[†]

WILEY

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To my parents

– *MNB*

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Preface to Third Edition

You have in front of you the third edition of the “Probability and Statistical Inference,” a text originally published in 1996. I have been using this book in the classroom since then, and it has always been interesting to see how it serves the students, how they react to it, and what could still be done to make it better. These reflections prompted me to prepare a second edition, published in 2007. But academia is changing quickly; who the students are is changing, and how we should teach to help them learn is changing as well. This is what made me consider a third edition. The response from Wiley Publishing was positive and my work began.

There were three main changes that I saw as necessary. First, adding a chapter on the basics of Bayesian statistics, as I realized that upper level undergraduate students and graduate students needed an earlier introduction to Bayesian inference. Another change was to make the book more appropriate for the flipped classroom format. I have experimented with it for three years now and it is working quite well. The book introduces and illustrates concepts through more than 400 examples. Preparing the material mainly at home gives students more time in class for questions, discussion, and for problem solving. I have also added over 70 new problems to make the selection easier for the instructor. A third change was including an appendix with an R code that would help students complete projects and homework assignments. My two-semester class based on this text includes three projects. The first one—in the fall semester—has students present applications of selected distributions, including graphics. Two projects for the spring semester involve resampling methods. The necessary R code is included in the appendix.

There are many people to whom I owe my thanks. First, I would like to thank Wiley Editor Jon Gurstelle, who liked the idea of preparing the third edition. After Jon accepted another job elsewhere, the book and I came under the excellent care of the Editorial Teams of Mindy Okura-Mokrzycki, Kathleen Santoloci, Linda Christina, and Kimberly Monroe-Hill who have supported me throughout this process. I would also like to thank Carla Koretsky, the Dean of the College of Arts and Sciences at Western Michigan University, and WMU Provost, Sue Stapleton, for granting me a semester-long administrative sabbatical leave that significantly sped up the progress of the book.

I am indebted to several of my students for their valuable comments. I am also grateful to my departmental colleagues, especially Hyun Bin Kang and Duy Ngo, who used the text in class and gave me their valuable feedback. Hyun Bin also helped me with the formatting of the R code in the appendix. Finally, I thank my husband, Jerzy, for his support and encouragement.

MNB

November 2020

Preface to Second Edition

The first edition of this book was published in 1996. Since then, powerful computers have come into wide use, and it became clear that our text should be revised and material on computer-intensive methods of statistical inference should be added. To my delight, Steve Quigley, Executive Editor of John Wiley and Sons, agreed with the idea, and work on the second edition began.

Unfortunately, Robert Bartoszyński passed away in 1998, so I was left to carry out this revision by myself. I revised the content by creating a new chapter on random samples, adding sections on Monte Carlo methods, bootstrap estimators and tests, and permutation tests. More problems were added, and existing ones were reorganized. Hopefully nothing was lost of the “spirit” of the book which Robert liked so much and of which he was very proud.

This book is intended for seniors or first-year graduate students in statistics, mathematics, natural sciences, engineering, and any other major where an intensive exposure to statistics is necessary. The prerequisite is a calculus sequence that includes multivariate calculus. We provide the material for a two-semester course that starts with the necessary background in probability theory, followed by the theory of statistics.

What distinguishes our book from other texts is the way the material is presented and the aspects that are stressed. To put it succinctly, understanding “why” is prioritized over the skill of “how to.” Today, in an era of undreamed-of computational facilities, a reflection in an attempt to understand is not a luxury but a necessity.

Probability theory and statistics are presented as self-contained conceptual structures. Their value as a means of description and inference about real-life situations lies precisely in their level of abstraction—the more abstract a concept is, the wider is its applicability. The methodology of statistics comes out most clearly if it is introduced as an abstract system illustrated by a variety of real-life applications, not confined to any single domain.

Depending on the level of the course, the instructor can select topics and examples, both in the theory and in applications. These can range from simple illustrations of concepts, to introductions of whole theories typically not included in comparable textbooks (e.g., prediction, extrapolation, and filtration in time series as examples of use of the concepts of covariance and correlation). Such additional, more advanced, material (e.g.,

Chapter 5 on Markov Chains) is marked with asterisks. Other examples are the proof of the extension theorem (Theorem 6.2.4), showing that the cumulative distribution function determines the measure on the line; the construction of Lebesgue, Riemann–Stieltjes, and Lebesgue–Stieltjes integrals; and the explanation of the difference between double integral and iterated integrals (Section 8.3).

In the material that is seldom included in other textbooks on mathematical statistics, we stress the consequences of nonuniqueness of a sample space and illustrate, by examples, how the choice of a sample space can facilitate the formulation of some problems (e.g., issues of selection or randomized response). We introduce the concept of conditioning with respect to partition (Section 4.4); we explain the Borel–Kolmogorov paradox by way of the underlying measurement process that provides information on the occurrence of the condition (Example 7.22); we present the Neyman–Scott theory of outliers (Example 10.4); we give a new version of the proof of the relation between mean, median, and standard deviation (Theorem 8.7.3); we show another way of conditioning in the secretary problem (Example 4.10). Among examples of applications, we discuss the strategies of serves in tennis (Problem 4.2.12), and a series of problems (3.2.14–3.2.20) concerning combinatorial analysis of voting power. In Chapter 11, we discuss the renewal paradox, the effects of importance sampling (Example 11.6), and the relevance of measurement theory for statistics (Section 11.6). Chapter 14 provides a discussion of true regression versus linear regression and concentrates mostly on explaining why certain procedures (in regression analysis and ANOVA) work, rather than on computational details. In Chapter 15, we provide a taste of rank methods—one line of research starting with the Glivenko–Cantelli Theorem and leading to Kolmogorov–Smirnov tests, and the other line leading to Mann–Whitney and Wilcoxon tests. In this chapter, we also show the traps associated with multiple tests of the same hypothesis (Example 15.3). Finally, Chapter 16 contains information on partitioning contingency tables—the method that provides insight into the dependence structure. We also introduce McNemar’s test and various indices of association for tables with ordered categories.

The backbone of the book is the examples used to illustrate concepts, theorems, and methods. Some examples raise the possibilities of extensions and generalizations, and some simply point out the relevant subtleties.

Another feature that distinguishes our book from most other texts is the choice of problems. Our strategy was to integrate the knowledge students acquired thus far, rather than to train them in a single skill or concept. The solution to a problem in a given section may require using knowledge from some preceding sections, that is, reaching back into material already covered. Most of the problems are intended to make the students aware of facts they might otherwise overlook. Many of the problems were inspired by our teaching experience and familiarity with students’ typical errors and misconceptions.

Finally, we hope that our book will be “friendly” for students at all levels. We provide (hopefully) lucid and convincing explanations and motivations, pointing out the difficulties and pitfalls of arguments. We also do not want good students to be left alone. The material in starred chapters, sections, and examples can be skipped in the main part of the course, but used at will by interested students to complement and enhance their knowledge. The book can also be a useful reference, or source of examples and problems, for instructors who teach courses from other texts.

I am indebted to many people without whom this book would not have reached its current form. First, thank you to many colleagues who contributed to the first edition and whose names are listed there. Comments of many instructors and students who used the first edition were influential in this revision. I wish to express my gratitude to Samuel Kotz for referring me to Stigler’s (1986) article about the “right and lawful rood,” which we previously used in the book without reference (Example 8.40). My sincere thanks are due to Jung Chao Wang

for his help in creating the R-code for computer-intensive procedures that, together with additional examples, can be found on the book's ftp site

ftp://ftp.wiley.com/public/sc_tech_med/probability_statistical.

Particular thanks are due to Katarzyna Bugaj for careful proofreading of the entire manuscript, Łukasz Bugaj for meticulously creating all figures with the Mathematica software, and Agata Bugaj for her help in compiling the index. Changing all those diapers has finally paid off.

I wish to express my appreciation to the anonymous reviewers for supporting the book and providing valuable suggestions, and to Steve Quigley, Executive Editor of John Wiley & Sons, for all his help and guidance in carrying out the revision.

Finally, I would like to thank my family, especially my husband Jerzy, for their encouragement and support during the years this book was being written.

Magdalena Niewiadomska-Bugaj

October 2007

About the Companion Website

This book is accompanied by a companion website:

www.wiley.com/go/probabilityandstatisticalinference3e

The website includes the Instructor's Solution Manual and will be live in early 2021.

CHAPTER 1

EXPERIMENTS, SAMPLE SPACES, AND EVENTS

1.1 INTRODUCTION

The consequences of making a decision today often depend on what will happen in the future, at least on the future that is relevant to the decision. The main purpose of using statistical methods is to help in making better decisions under uncertainty.

Judging from the failures of weather forecasts, to more spectacular prediction failures, such as bankruptcies of large companies and stock market crashes, it would appear that statistical methods do not perform very well. However, with a possible exception of weather forecasting, these examples are, at best, only partially statistical predictions. Moreover, failures tend to be better remembered than successes. Whatever the case, statistical methods are at present, and are likely to remain indefinitely, our best and most reliable prediction tools.

To make decisions under uncertainty, one usually needs to collect some *data*. Data may come from past experiences and observations, or may result from some controlled processes, such as laboratory or field experiments. The data are then used to hypothesize about the laws (often called *mechanisms*) that govern the fragment of reality of interest. In our book, we are interested in laws expressed in probabilistic terms: They specify directly, or allow us to compute, the chances of some events to occur. Knowledge of these chances is, in most cases, the best one can get with regard to prediction and decisions.

Probability theory is a domain of pure mathematics and as such, it has its own conceptual structure. To enable a variety of applications (typically comprising of all areas of human endeavor, ranging from biological, medical, social and physical sciences, to engineering, humanities, business, etc.), such structure must be kept on an abstract level. An application of probability to the particular situation analyzed requires a number of initial steps in which the elements of the real situation are *interpreted* as abstract concepts of probability theory.

Such interpretation is often referred to as building a *probabilistic model* of the situation at hand. How well this is done is crucial to the success of application.

One of the main concepts here is that of an *experiment*—a term used in a broad sense. It means any process that generates data which is influenced, at least in part, by chance.

1.2 SAMPLE SPACE

In analyzing an experiment, one is primarily interested in its *outcome*—the concept that is not defined (i.e., a *primitive concept*) but has to be specified in every particular application. This specification may be done in different ways, with the only requirements being that (1) outcomes exclude one another and (2) they exhaust the set of all logical possibilities.

■ EXAMPLE 1.1

Consider an experiment consisting of two tosses of a regular die. An outcome is most naturally represented by a pair of numbers that turn up on the upper faces of the die so that they form a pair (x, y) , with $x, y = 1, 2, \dots, 6$ (see Table 1.1).

Table 1.1 Outcomes on a pair of dice.

		y					
		1	2	3	4	5	6
x	1	(1, 1)	(1, 2)	(1, 3)	(1, 4)	(1, 5)	(1, 6)
	2	(2, 1)	(2, 2)	(2, 3)	(2, 4)	(2, 5)	(2, 6)
	3	(3, 1)	(3, 2)	(3, 3)	(3, 4)	(3, 5)	(3, 6)
	4	(4, 1)	(4, 2)	(4, 3)	(4, 4)	(4, 5)	(4, 6)
	5	(5, 1)	(5, 2)	(5, 3)	(5, 4)	(5, 5)	(5, 6)
	6	(6, 1)	(6, 2)	(6, 3)	(6, 4)	(6, 5)	(6, 6)

In the case of an experiment of tossing a die three times, the outcomes will be triplets (x, y, z) , with x, y , and z being integers between 1 and 6.

Since the outcome of an experiment is not known in advance, it is important to determine the set of all possible outcomes. This set, called the *sample space*, forms the conceptual framework for all further considerations of probability.

Definition 1.2.1 The *sample space*, denoted by $\mathcal{S}$, is the set of all outcomes of an experiment. The elements of the sample space are called *elementary outcomes*, or *sample points*. $\square$

■ EXAMPLE 1.2

In Example 1.1, the sample space $\mathcal{S}$ has $6^2 = 36$ sample points in the case of two tosses, and $6^3 = 216$ points in the case of three tosses of a die. The first statement can be verified by a direct counting of the elements of the sample space. Similar verification of the second claim, although possible in principle, would be cumbersome. In Chapter 3, we will introduce some methods of determining the sizes of sets without actually counting sample points.

■ EXAMPLE 1.3

Suppose that the only available information about the numbers, those that turn up on the upper faces of the die, is their sum. In such a case as outcomes, we take 11 possible values of the sum so that

$$\mathcal{S} = \{2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}.$$

For instance, all outcomes on the diagonal of Table 1.1—(6, 1), (5, 2), (4, 3), (3, 4), (2, 5), and (1, 6)—are represented by the same value 7.

■ EXAMPLE 1.4

If we are interested in the number of accidents that occur at a given intersection within a month, the sample space might be taken as the set $\mathcal{S} = \{0, 1, 2, \dots\}$ consisting of all nonnegative integers. Realistically, there is a practical limit, say 1,000, of the monthly numbers of accidents at this particular intersection. Although one may think that it is simpler to take the sample space $\mathcal{S} = \{0, 1, 2, \dots, 1,000\}$, it turns out that it is often much simpler to take the infinite sample space if the “practical bound” is not very precise.

Since outcomes can be specified in various ways (as illustrated by Examples 1.1 and 1.3), it follows that the same experiment can be described in terms of different sample spaces $\mathcal{S}$. The choice of a sample space depends on the goal of description. Moreover, certain sample spaces for the same experiment lead to easier and simpler analysis. The choice of a “better” sample space requires some skill, which is usually gained through experience. The following two examples illustrate this point.

■ EXAMPLE 1.5

Let the experiment consist of recording the lifetime of a piece of equipment, say a light bulb. An outcome here is the time until the bulb burns out. An outcome typically will be represented by a number $t \geq 0$ ($t = 0$ if the bulb is not working at the start), and therefore $\mathcal{S}$ is the nonnegative part of the real axis. In practice, t is measured with some precision (in hours, days, etc.), so one might instead take $\mathcal{S} = \{0, 1, 2, \dots\}$. Which of these choices is better depends on the type of subsequent analysis.

■ EXAMPLE 1.6

Two persons enter a cafeteria and sit at a square table, with one chair on each of its sides. Suppose we are interested in the event “they sit at a corner” (as opposed to sitting across from one another). To construct the sample space, we let A and B denote the two persons, and then take as $\mathcal{S}$ the set of outcomes represented by 12 ideograms in Figure 1.1. One could argue, however, that such a sample space is unnecessarily large.

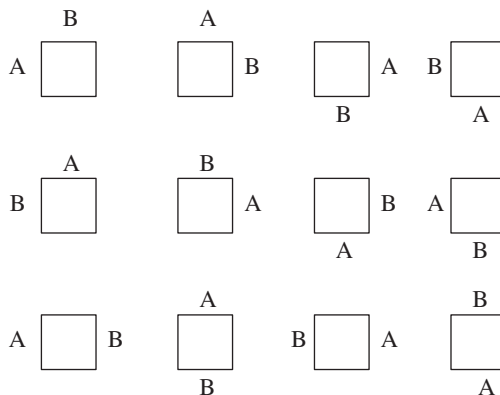


Figure 1.1 Possible seatings of persons A and B at a square table.

If we are interested only in the event “they sit at a corner,” then there is no need to label the persons as A and B. Accordingly, the sample space S may be reduced to the set of six outcomes depicted in Figure 1.2. But even this sample space can be simplified. Indeed, one could use the rotational symmetry of the table and argue that once the first person selects a chair (it does not matter which one), then the sample space consists of just three chairs remaining for the second person (see Figure 1.3).

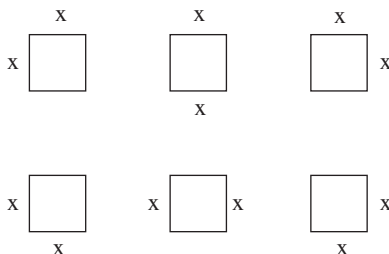


Figure 1.2 Possible seatings of any two persons at a square table.

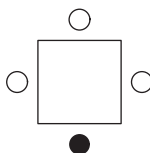


Figure 1.3 Possible seatings of one person if the place of the other person is fixed.

Sample spaces can be classified according to the number of sample points they contain. *Finite* sample spaces contain finitely many outcomes, and elements of *infinitely countable* sample spaces can be arranged into an infinite sequence; other sample spaces are called *uncountable*.

The next concept to be introduced is that of an *event*. Intuitively, an event is anything about which we can tell whether or not it has occurred, as soon as we know the outcome of the experiment. This leads to the following definition:

Definition 1.2.2 An *event* is a subset of the sample space S . □

■ EXAMPLE 1.7

In Example 1.1 an event such as “the sum equals 7” containing six outcomes $(1, 6), (2, 5), (3, 4), (4, 3), (5, 2),$ and $(6, 1)$ is a subset of the sample space S . In Example 1.3, the same event consists of one outcome, 7.

When an experiment is performed, we observe its outcome. In the interpretation developed in this chapter, this means that we observe a point chosen randomly from the sample space. If this point belongs to the subset representing the event A , we say that *the event A has occurred*.

We will let events be denoted either by letters $A, B, C, \dots$, possibly with identifiers, such as $A_1, B_k, \dots$, or by more descriptive means, such as $\{X = 1\}$ and $\{a < Z < b\}$, where X

and Z are some numerical attributes of the sample points (formally: random variables, to be discussed in Chapter 5). Events can also be described through verbal phrases, such as “two heads in a row occur before the third tail” in the experiment of repeated tosses of a coin.

In all cases considered thus far, we assumed that an outcome (a point in the sample space) *can be observed*. To put it more precisely, all sample spaces $\mathcal{S}$ considered so far were constructed in such a way that their points were observable. Thus, for any event A , we were always able to tell whether it occurred or not.

The following examples show experiments and corresponding sample spaces with sample points that are only *partially observable*:

■ EXAMPLE 1.8 Selection

Candidates for a certain job are characterized by their level z of skills required for the job. The actual value of z is not observable, though; what we observe is the candidate's score x on a certain test. Thus, the sample point in $\mathcal{S}$ is a pair $s = (z, x)$, and only one coordinate of s , x , is observable.

The objective might be to find selection thresholds z_0 and x_0 , such that the rule: “accept all candidates whose score x exceeds x_0 ” would lead to maximizing the (unobservable) number of persons accepted whose true level of skill z exceeds z_0 . Naturally, to find such a solution, one needs to understand statistical relation between observable x and unobservable z .

Another example when the points in the sample space are only partially observable concerns studies of incidence of activities about which one may hesitate to respond truthfully, or even to respond at all. These are typically studies related to sexual habits or preferences, abortion, law and tax violation, drug use, and so on.

■ EXAMPLE 1.9 Randomized Response

Let Q be the activity analyzed, and assume that the researcher is interested in the frequency of persons who ever participated in activity Q (for simplicity, we will call them Q -persons). It ought to be stressed that the objective is *not* to identify the Q -persons, but only to find the proportion of such persons in the population.

The direct question reduced to something like “Are you a Q -person?” is not likely to be answered truthfully, if at all. It is therefore necessary to make the respondent safe, guaranteeing that their responses will reveal nothing about them as regards Q . This can be accomplished as follows: The respondent is given a pair of distinguishable dice, for example, one green and one white. She throws them both at the same time, in such a way that the experimenter does not know the results of the toss (e.g., the dice are in a box and only the respondent looks into the box after it is shaken). The instruction is the following: If the green die shows an odd face (1, 3, or 5), then respond to the question “Are you a Q -person?” If the green die shows an even face (2, 4, or 6), then respond to the question, “Does the white die show an ace?” The scheme of this response is summarized by the flowchart in Figure 1.4.

The interviewer knows the answer “yes” or “no” but does not know whether it is the answer to the question about Q or the question about the white die. Here a natural sample space consists of points $s = (i, x, y)$, where x and y are outcomes on green and white die, respectively, while i is 1 or 0 depending on whether or not the respondent is a Q -person. We have $\phi(s) = \phi(i, x, y) =$ “yes” if $i = 1$ and $x = 1, 3$, or 5 for any y , or if $x = 2, 4, 6$, and $y = 1$ for any i . In all other cases, $\phi(s) =$ “no.”

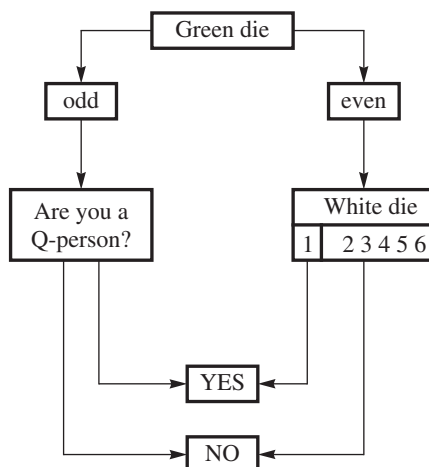


Figure 1.4 Scheme of a randomized response.

One could wonder what is a possible advantage, if any, of not knowing the question asked and observing only the answer. This does not make sense if we need to know the truth about each individual respondent. However, one should remember that we are only after the overall frequency of Q -persons.

We are in fact “contaminating” the question by making the respondent answer either a Q -question or some other auxiliary question. But this is a “controlled contamination”: we know how often (on average) the respondents answer the auxiliary question, and how often their answer is “yes.” Consequently, as we will see in Chapter 11, we can still make an inference about the proportion of Q -persons based on the observed responses.

PROBLEMS

- 1.2.1** List all sample points in sample spaces for the following experiments: **(i)** We toss a balanced coin.¹ If heads come up, we toss a die. Otherwise, we toss the coin two more times. **(ii)** A coin is tossed until the total of two tails occurs, but no more than four times (i.e., a coin is tossed until the second tail or fourth toss, whichever comes first).
- 1.2.2** Alice, Bob, Carl, and Diana enter the elevator on the first floor of a four-story building. Each of them leaves the elevator on either the second, third, or fourth floor. **(i)** Describe the sample space without listing all sample points. **(ii)** List all sample points such that Carl and Diana leave the elevator on the third floor. **(iii)** List all sample points if Carl and Diana leave the elevator at the same floor.
- 1.2.3** An urn contains five chips, labeled 1, $\dots$, 5. Three chips are selected. List all outcomes included in the event “the second largest number drawn was 3.”
- 1.2.4** In a game of craps, the player rolls a pair of dice. If he gets a total of 7 or 11, he wins at once; if the total is 2, 3, or 12, he loses at once. Otherwise, the sum, say x , is his “point,” and he keeps rolling dice until either he rolls another x (in which case he wins) or he rolls a 7 in which case he loses. Describe the event “the player wins with a point of 5.”

¹Unless specifically stated, we will be assuming that all coins and/or dice tossed are fair (balanced).

- 1.2.5** The experiment consists of placing six balls in three boxes. List all outcomes in the sample space if: **(i)** The balls are indistinguishable, but the boxes are distinguishable. (*Hint:* There are 28 different placements.) **(ii)** Neither the balls nor the boxes are distinguishable. **(iii)** Two balls are white and four are red; the boxes are distinguishable.
- 1.2.6** John and Mary plan to meet each other. Each of them is to arrive at the meeting place at some time between 5 p.m. and 6 p.m. John is to wait 20 minutes (or until 6 p.m., whichever comes first), and then leave if Mary does not show up. Mary will wait only 5 minutes (or until 6 p.m., whichever comes first), and then leave if John does not show up. Letting x and y denote the arrival times of John and Mary, determine the sample space and describe events (i)–(viii) by drawing pictures, or by appropriate inequalities for x and y . If you think that the description is impossible, say so. **(i)** John arrives before Mary does. **(ii)** John and Mary meet. **(iii)** Either Mary comes first, or they do not meet. **(iv)** Mary comes first, but they do not meet. **(v)** John comes very late. **(vi)** They arrive less than 15 minutes apart, and they do not meet. **(vii)** Mary arrives at 5:15 p.m. and meets John, who is already there. **(viii)** They almost miss one another.

Problems 1.2.7–1.2.8 show an importance of the sample space selection.

- 1.2.7** Let $\mathcal{E}$ be the experiment consisting of tossing a coin three times, with H and T standing for heads and tails, respectively.

(i) The following set of outcomes is an incomplete list of the points of the sample space $\mathcal{S}$ of the experiment $\mathcal{E}$: {HHH, HTT, TTT, HHT, HTH, THH}. Use a tree diagram to find the missing outcomes.

(ii) An alternative sample space $\mathcal{S}'$ for the same experiment $\mathcal{E}$ consists of the following four outcomes: no heads (0), one head (1), two heads (2), and three heads (3). Which of the following events can be described as subsets of $\mathcal{S}$ but not as subsets of $\mathcal{S}' = \{0, 1, 2, 3\}$?

A_1 = More than two heads.

A_2 = Head on the second toss.

A_3 = More tails than heads.

A_4 = At least one tail, with head on the last toss.

A_5 = At least two faces the same.

A_6 = Head and tail alternate.

(iii) Still another sample space $\mathcal{S}''$ for the experiment $\mathcal{E}$ consists of the four outcomes (0, 0), (0, 1), (1, 0), and (1, 1). The first coordinate is 1 if the first two tosses show the same face and 0 otherwise; the second coordinate is 1 if the last two tosses show the same face, and 0 otherwise. For instance, if we observe HHT, the outcome is (1, 0). List the outcomes of $\mathcal{S}$ that belong to the event $A = \{(1, 1), (0, 1)\}$ of $\mathcal{S}''$.

(iv) Which of the following events can be represented as subsets of $\mathcal{S}$, but cannot be represented as subsets of $\mathcal{S}''$?

B_1 = First and third toss show the same face.

B_2 = Heads on all tosses.

B_3 = All faces the same.

B_4 = Each face appears at least once.

B_5 = More heads than tails.

- 1.2.8** Let $\mathcal{E}$ be an experiment consisting of tossing a die twice. Let $\mathcal{S}$ be the sample space with sample points (i, j) , $i, j = 1, 2, \dots, 6$, with i and j being the numbers of dots that appear in the first and second toss, respectively.

(i) Let S' be the sample space for the experiment $\mathcal{E}$ consisting of all possible sums $i + j$ so that $S' = \{2, 3, \dots, 12\}$. Which of the following events can be defined as subsets of $\mathcal{S}$ but not of S' ?

A_1 = One face odd, the other even.

A_2 = Both faces even.

A_3 = Faces different.

A_4 = Result on the first toss less than the result on the second.

A_5 = Product greater than 10.

A_6 = Product greater than 30.

(ii) Let S'' be the sample space for the experiment $\mathcal{E}$ consisting of all possible absolute values of the difference $|i - j|$ so that $S'' = \{0, 1, 2, 3, 4, 5\}$. Which of the following events can be defined as subsets of $\mathcal{S}$ but not of S'' ?

B_1 = One face shows twice as many dots as the other,

B_2 = Faces the same,

B_3 = One face shows six times as many dots as the other,

B_4 = One face odd, the other even,

B_5 = The ratio of the numbers of dots on the faces is different from 1.

1.2.9 Referring to Example 1.9, suppose that we modify it as follows: The respondent tosses a green die (with the outcome unknown to the interviewer). If the outcome is odd, he responds to the Q-question; otherwise, he responds to the question “Were you born in April?” Again, the interviewer observes only the answer “yes” or “no.” Apart from the obvious difference in the frequency of the answer “yes” to the auxiliary question (on the average 1 in 12 instead of 1 in 6), are there any essential differences between this scheme and the scheme in Example 1.9? Explain your answer.

1.3 ALGEBRA OF EVENTS

Next, we introduce concepts that will allow us to form composite events out of simpler ones. We begin with the relations of *inclusion* and *equality*.

Definition 1.3.1 The event A is *contained* in the event B , or B *contains* A , if every sample point of A is also a sample point of B . Whenever this is true, we will write $A \subset B$, or equivalently, $B \supset A$. $\square$

An alternative terminology here is that A *implies* (or *entails*) B .

Definition 1.3.2 Two events A and B are said to be *equal*, $A = B$, if $A \subset B$ and $B \subset A$. $\square$

It follows that two events are equal if they consist of exactly the same sample points.

■ EXAMPLE 1.10

Consider two tosses of a coin, and the corresponding sample space $\mathcal{S}$ consisting of four outcomes: HH, HT, TH, and TT. The event A = “heads in the first toss” = {HH, HT} is contained in the event B = “at least one head” = {HH, HT, TH}. The events “the results alternate” and “at least one head and one tail” imply one another, and hence are equal.

Definition 1.3.3 The set containing no elements is called the *empty* set and is denoted by $\emptyset$. The event corresponding to $\emptyset$ is called a *null (impossible)* event. $\square$

■ EXAMPLE 1.11 *²

The reader may wonder whether it is correct to use the definite article in the definition above and speak of “*the* empty set,” since it would appear that there may be many different empty sets. For instance, the set of all kings of the United States and the set of all real numbers x such that $x^2 + 1 = 0$ are both empty, but one consists of people and the other of numbers, so they cannot be equal. This is not so, however, as is shown by the following formal argument (to appreciate this argument, one needs some training in logic). Suppose that $\emptyset_1$ and $\emptyset_2$ are two empty sets. To prove that they are equal, one needs to prove that $\emptyset_1 \subset \emptyset_2$ and $\emptyset_2 \subset \emptyset_1$. Formally, the first inclusion is the implication: “if s belongs to $\emptyset_1$, then s belongs to $\emptyset_2$.” This implication is true, because its premise is false: there is no s that belongs to $\emptyset_1$. The same holds for the second implication, so $\emptyset_1 = \emptyset_2$.

We now give the definitions of three principal operations on events: *complementation*, *union*, and *intersection*.

Definition 1.3.4 The set that contains all sample points that are not in the event A will be called the *complement* of A and denoted A^c , to be read also as “not A .” $\square$

Definition 1.3.5 The set that contains all sample points belonging either to A or to B (so possibly to both of them) is called the *union* of A and B and denoted $A \cup B$, to be read as “ A or B .” $\square$

Definition 1.3.6 The set that contains all sample points belonging to both A and B is called the *intersection* of A and B and denoted $A \cap B$. $\square$

An alternative notation for a complement is A' or $\overline{A}$, whereas in the case of an intersection, one often writes AB instead of $A \cap B$.

The operations above have the following interpretations in terms of occurrences of events:

1. Event A^c occurs if event A does not occur.
2. Event $A \cup B$ occurs when either A or B or both events occur.
3. Event $A \cap B$ occurs when both A and B occur.

■ EXAMPLE 1.12

Consider an experiment of tossing a coin three times, with the sample space consisting of outcomes described as HHH, HHT, and so on. Let A and B be the events “heads and tails alternate” and “heads on the last toss,” respectively. The event A^c occurs if either heads or tails occur at least twice in a row so that $A^c = \{HHH, HHT, THH, HTT, TTT, TTH\}$, while B^c is “tails on the last toss,” hence, $B^c = \{HHT, THT, HTT, TTT\}$. The union $A \cup B$ is the event “either the results alternate or it is heads on the last toss,” meaning $A \cup B = \{HTH, THT, HHH, THH, TTH\}$. Observe that while A has two outcomes and B has

²Asterisks denote more advanced material, as explained in the Preface to the Second Edition.

four outcomes, their union has only five outcomes, since the outcome HTH appears in both events. This common part is the intersection $A \cap B$.

Some formulas can be simplified by introducing the operation of the *difference* of two events.

Definition 1.3.7 The *difference*, $A \setminus B$, of events A and B contains all sample points that belong to A but not to B

$$A \setminus B = A \cap B^c.$$

The *symmetric difference*, $A \div B$, contains sample points that belong to A or to B , but not to both of them:

$$A \div B = (A \cap B^c) \cup (A^c \cap B) = (A \cup B) \setminus (A \cap B). \quad \square$$

■ EXAMPLE 1.13

In Example 1.12, the difference $B^c \setminus A$ is described as “at least two identical outcomes in a row and tails on the last toss,” which means the event $\{\text{HHT}, \text{HTT}, \text{TTT}\}$.

Next, we introduce the following important concept:

Definition 1.3.8 If $A \cap B = \emptyset$, then the events A and B are called *disjoint*, or *mutually exclusive*. $\square$

■ EXAMPLE 1.14

Based on Example 1.12, we know that the following two events are disjoint: C = “more heads than tails” and the intersection $A \cap B^c$ = “the results alternate, ending with tails.”

Example 1.14 shows that to determine whether or not events are disjoint, it is not necessary to list the outcomes in both events and check whether there exist common outcomes. Apart from the fact that such listing is not feasible when sample spaces are large, it is often simpler to employ logical reasoning. In the case above, if the results alternate and end with tails, then the outcome must be THT. Since there are more tails than heads, C does not occur.

The definitions of union and intersection can be extended to the case of a finite and even infinite number of events (to be discussed in the Section 1.4). Thus,

$$A_1 \cup A_2 \cup \cdots \cup A_n = \bigcup_{i=1}^n A_i \quad (1.1)$$

is the event that contains the sample points belonging to A_1 or A_2 or $\dots$ or A_n . Consequently, (1.1) is the event “at least one A_i occurs.” Similarly,

$$A_1 \cap A_2 \cap \cdots \cap A_n = \bigcap_{i=1}^n A_i \quad (1.2)$$

is the event that contains the sample points belonging to A_1 and A_2 and $\dots$ and A_n . Consequently, the event (1.2) is “all A_i ’s occur.”

■ EXAMPLE 1.15

Suppose that n shots are fired at a target, and let A_i , $i = 1, 2, \dots, n$ denote the event “the target is hit on the i th shot.” Then, the union $A_1 \cup \cdots \cup A_n$ is the event “the

target is hit” (at least once). Its complement $(A_1 \cup \dots \cup A_n)^c$ is the event “the target is missed” (on every shot), which is the same as the intersection $A_1^c \cap \dots \cap A_n^c$.

A perceptive reader may note that the unions $A_1 \cup \dots \cup A_n$ and intersections $A_1 \cap \dots \cap A_n$ do not require an extension of the definition of union and intersection for two events. Indeed, we could consider unions such as

$$A_1 \cup (A_2 \cup (\dots (A_{n-2} \cup (A_{n-1} \cup A_n)) \dots)),$$

where the union of only two events is formed in each set of parentheses. The property of associativity (below) shows that parentheses can be omitted so that the expression $A_1 \cup \dots \cup A_n$ is unambiguous. The same argument applies to intersections.

The operations on events defined in this section obey some laws. The most important ones are listed below.

Idempotence:

$$A \cup A = A, \quad A \cap A = A.$$

Double complementation:

$$(A^c)^c = A.$$

Absorption:

$$A \cup B = B \iff A \cap B = A \iff A \subset B. \quad (1.3)$$

In particular,

$$A \cup \emptyset = A, \quad A \cup \mathcal{S} = \mathcal{S}, \quad A \cap \emptyset = \emptyset, \quad A \cap \mathcal{S} = A,$$

which in view of (1.3) means that $\emptyset \subset A \subset \mathcal{S}$.

Commutativity:

$$A \cup B = B \cup A, \quad A \cap B = B \cap A.$$

Associativity:

$$A \cup (B \cup C) = (A \cup B) \cup C, \quad A \cap (B \cap C) = (A \cap B) \cap C.$$

Distributivity:

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C), \quad A \cup (B \cap C) = (A \cup B) \cap (A \cup C).$$

De Morgan's Laws:

$$\begin{aligned} (A_1 \cup \dots \cup A_n)^c &= A_1^c \cap \dots \cap A_n^c, \\ (A_1 \cap \dots \cap A_n)^c &= A_1^c \cup \dots \cup A_n^c. \end{aligned} \quad (1.4)$$

It is often helpful to use Venn diagrams for studying relations between composite events in the same sample space. The sample space $\mathcal{S}$ is there represented by a rectangle, while its subsets represent events (see Figure 1.5).

The complement of event A is represented in Figure 1.5a, the union and intersection of the events A and B are represented in Figure 1.5b and c, respectively.

Venn diagrams can also be used to check the validity of formulas. For example, consider the first De Morgan's law (1.4) for the case of two events:

$$(A \cup B)^c = A^c \cap B^c. \quad (1.5)$$

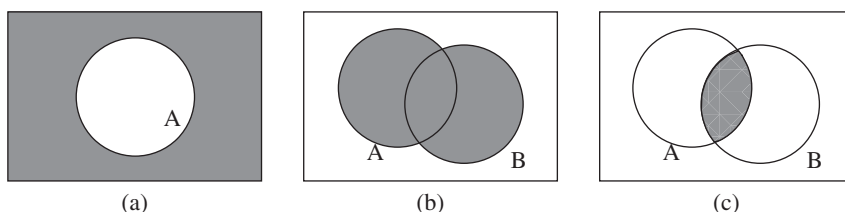


Figure 1.5 Complement, union, and intersection.

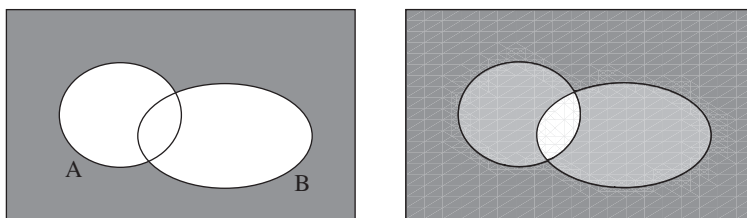


Figure 1.6 The first De Morgan's law.

Venn diagrams made separately for the left-hand side and the right-hand side of (1.5) (see Figure 1.6) indicate that both regions are the same. Although a picture does not constitute a proof, it may provide convincing evidence that the statement is true, and sometimes may even suggest a method of proving the statement.

PROBLEMS

For the problems below, remember that a statement (expressed as a sentence or formula) is true if it is true under *all* circumstances, and it is false if there is at least one case where it does not hold.

1.3.1 Answer true or false. Justify your answer. **(i)** If A and B are distinct events (i.e., $A \neq B$) such that A and B^c are disjoint, then A^c and B are also disjoint. **(ii)** If A and B are disjoint, then A^c and B^c are also disjoint. **(iii)** If A and B are disjoint, and also B and C are disjoint, then A and C are disjoint. **(iv)** If A and B are both contained in C , then $C^c \subset A^c \cap B^c$. **(v)** If A is contained in B , C is contained in D , and B is disjoint from D , then A is disjoint from C . **(vi)** If $A \cup B^c = B^c$, then $B \subset A^c$.

1.3.2 In the statements below A, B, C , and D are events. Find those statements or formulas that are true. **(i)** If $A \cap B = A \cap C$, then $B = C$. **(ii)** $A \cup (A \cap B) \cup (A \cap B^c) = A$. **(iii)** $A \cup (A \cap B) \cup (A \cap B^c) = B$. **(iv)** If $A \setminus B = C$, then $A = B \cup C$. **(v)** $(A \cup B) \cap (C \cup D) = (A \cap C) \cup (A \cap D) \cup (B \cap C) \cup (B \cap D)$. **(vi)** $(A \cap B) \cup (C \cap D) = (A \cup C) \cap (A \cup D) \cap (B \cup C) \cap (B \cup D)$. **(vii)** $(A^c \cup B^c \cup C^c)^c = A^c \cap B^c \cap C^c$. **(viii)** If $A \subset B$, and $B \cap C = \emptyset$, then $C^c \cap A \cap B^c = \emptyset$. **(ix)** If $A \cap B$, $A \cap C$ and $B \cap C$ are not empty, then $A \cap B \cap C$ is not empty. **(x)** Show that $(A \div B) \div C = A \div (B \div C)$.

1.3.3 Find X if: **(i)** $A \div X = \emptyset$. **(ii)** $A \div X = A$. **(iii)** $A \div X = S$. **(iv)** $A \div X = B$.

1.3.4 In a group of 1,000 students of a certain college, 60 take French, 417 take calculus, and 509 take statistics. Moreover, 20 take French and calculus, 17 take French and statistics, and 147 take statistics and calculus. However, 196 students do not take any

of these three subjects. Determine the number of students who take French, calculus, and statistics.

- 1.3.5** Let A , B , and C be three events. Match, where possible, events D_1 through D_{10} with events E_1 through E_{11} . Matching means that the events are exactly the same; that is, if one occurs, so must the other and conversely (see the Definition 1.3.2). (*Hint*: Draw a Venn diagram for each event $D_1, \dots, D_{10}$, do the same for events $E_1, \dots, E_{11}$, and then compare the diagrams.)

Among events A, B, C :

- | | |
|---------------------------------|-----------------------------|
| D_1 = two or more occur. | D_2 = exactly one occurs. |
| D_3 = only A occurs. | D_4 = all occur. |
| D_5 = none occurs. | D_6 = at most one occurs. |
| D_7 = at least one occurs. | D_8 = exactly two occur. |
| D_9 = no more than two occur. | D_{10} = B occurs. |

$$\begin{aligned}
 E_1 &= A \cup B \cup C. \\
 E_2 &= (A \cap B^c \cap C^c) \cup (A^c \cap B \cap C^c) \cup (A^c \cap B^c \cap C). \\
 E_3 &= (A \cap B)^c \cap (A \cap C)^c \cap (B \cap C)^c. \\
 E_4 &= (A \cup B \cup C)^c. \\
 E_5 &= A^c \cap B^c \cap C^c. \\
 E_6 &= A \cap B \cap C. \\
 E_7 &= B. \\
 E_8 &= A \cap B^c \cap C^c. \\
 E_9 &= (A \cap B \cap C^c) \cup (A \cap B^c \cap C) \cup (A^c \cap B \cap C). \\
 E_{10} &= (A \cap B \cap C)^c. \\
 E_{11} &= (A \cap B) \cup (A \cap C) \cup (B \cap C).
 \end{aligned}$$

- 1.3.6** A standard deck of cards is dealt among players N, S, E , and W . Let $N_k, k = 1, 2, 3, 4$ be the event “ N has at least k aces,” and let S_k, E_k , and W_k be defined similarly. For each of the events below, determine the number of aces that N has. **(i)** $N_1 \cap S_1 \cap E_1 \cap W_1$. **(ii)** $E_2 \cap (W_2 \cup S_2)$. **(iii)** $N_3 \setminus N_4$. **(iv)** $S_3 \cap W_1$. **(v)** $S_1^c \cap W_1^c \cap E_1^c$. **(vi)** $N_2 \cap E_2$.
- 1.3.7** Five burglars, A, B, C, D , and E , divide the loot, consisting of five identical gold bars and four identical diamonds. Let A_{jk} be the event that A got *at least* j gold bars and *at most* k diamonds. Let B_{jk}, C_{jk} denote analogous events for burglars B, C (e.g., B_{21} is the event that B got 2, 3, 4, or 5 gold bars and 0 or 1 diamond). Determine the number x of gold bars and the number y of diamonds received by E if the following events occur (if determination of x and/or y is impossible, give the range of values): **(i)** $(A_{20} \cup B_{20} \cup C_{20}) \cap D_{30}$. **(ii)** E_{12}^c . **(iii)** $A_{23} \cap B_{13} \cap C_{13} \cap D_{13}$. **(iv)** $A_{23} \cup B_{13} \cup C_{13} \cup D_{13}$.
- 1.3.8** Let A^{nc} be defined inductively by $A^{0c} = A, A^{(n+1)c} = (A^{nc})^c$. Find $A^{mc} \cap A^{nc}$ and $A^{mc} \cup A^{nc}$ for $m, n > 0$.

1.4 INFINITE OPERATIONS ON EVENTS

As already mentioned, the operations of union and intersection can be extended to infinitely many events. Let $A_1, A_2, \dots$ be an infinite sequence of events. Then,

$$A_1 \cup A_2 \cup \dots = \bigcup_{i=1}^{\infty} A_i \quad \text{and} \quad A_1 \cap A_2 \cap \dots = \bigcap_{i=1}^{\infty} A_i$$

are events “at least one A_i occurs” and “all A_i ’s occur,” respectively.

If at least one event A_i occurs, then there is one that occurs first. This remark leads to the following useful decomposition of a union of events into a union of *disjoint* events:

$$\bigcup_{i=1}^{\infty} A_i = A_1 \cup (A_1^c \cap A_2) \cup (A_1^c \cap A_2^c \cap A_3) \cup \cdots, \quad (1.6)$$

where $A_1^c \cap \cdots \cap A_{k-1}^c \cap A_k$ is the event “ A_k is the first event in the sequence that occurs.”

For an infinite sequence $A_1, A_2, \dots$ one can define two events:

$$\limsup A_n = \bigcap_{k=1}^{\infty} \bigcup_{i=k}^{\infty} A_i \quad (1.7)$$

and

$$\liminf A_n = \bigcup_{k=1}^{\infty} \bigcap_{i=k}^{\infty} A_i, \quad (1.8)$$

these being, respectively, the event that “infinitely many A_i ’s occur” and the event that “all except finitely many A_i ’s occur.” Here the inner union in the event (1.7) is the event “at least one event A_i with $i \geq k$ will occur”; call this event B_k . The intersection over k means that the event B_k occurs for every k . No matter how large k we take, there will be at least one event A_i with $i \geq k$ that will occur. But this is possible only if infinitely many A_i s occur.

For the event $\liminf A_n$, the argument is similar. The intersection $A_k \cap A_{k+1} \cap \cdots = C_k$ occurs if all events A_i with $i \geq k$ occur. The union $C_1 \cup C_2 \cup \cdots$ means that at least one of the events C_k will occur, and that means that all A_i will occur, except possibly finitely many.

If all events (except possibly finitely many) occur, then infinitely many of them must occur, so that $\limsup A_n \supset \liminf A_n$. If $\limsup A_n \subset \liminf A_n$, then (see the definition of equality of events) we say that the sequence $\{A_n\}$ *converges*, and $\limsup A_n = \liminf A_n$.

The most important class of convergent sequences of events consists of *monotone* sequences, when $A_1 \subset A_2 \subset \cdots$ (increasing sequence) or when $A_1 \supset A_2 \supset \cdots$ (decreasing sequence). We have the following theorem:

Theorem 1.4.1 *If the sequence $A_1, A_2, \dots$ is increasing, then*

$$\lim A_n = \bigcup_{n=1}^{\infty} A_n,$$

and in case of a decreasing sequence, we have

$$\lim A_n = \bigcap_{n=1}^{\infty} A_n.$$

Proof: If the sequence is increasing, then the inner union $(\bigcup_{i=1}^{\infty} A_i)$ in $\limsup A_n$ remains the same independently of k so that $\limsup A_n = \bigcup_{i=1}^{\infty} A_i$. On the other hand, the inner intersection in $\liminf A_n$ equals A_k so that $\liminf A_n = \bigcup_{k=1}^{\infty} A_k$, which is the same as $\limsup A_n$, as was to be shown. A similar argument holds for decreasing sequences. $\square$

The following two examples illustrate the concept of convergence of events.

■ EXAMPLE 1.16

Let $B(r)$ and $C(r)$ be the sets of points on the plane (x, y) satisfying the conditions $x^2 + y^2 < r^2$ and $x^2 + y^2 \leq r^2$, respectively. If $A_n = B(1 + 1/n)$, then $\{A_n\}$ is a decreasing sequence, and therefore $\lim A_n = \bigcap_{n=1}^{\infty} B(1 + 1/n)$. Since $x^2 + y^2 <$

$(1 + 1/n)^2$ for all n if and only if $x^2 + y^2 \leq 1$, we have $\bigcap_{n=1}^{\infty} B(1 + 1/n) = C(1)$. On the other hand, if $A_n = C(1 - 1/n)$, then $\{A_n\}$ is an increasing sequence, and $\lim A_n = \bigcup_{n=1}^{\infty} C(1 - 1/n) = B(1)$. We leave a justification of the last equality to the reader.

■ EXAMPLE 1.17

Let $A_n = B(1 + 1/n)$ for n odd and $A_n = B(1/3 - 1/2n)$ for n even. The sequence $\{A_n\}$ is now $B(2), B(1/12), B(4/3), B(5/24), \dots$, so it is not monotone. We have here $\limsup A_n = C(1)$, since every point (x, y) with $x^2 + y^2 \leq 1$ belongs to infinitely many A_n . On the other hand, $\liminf A_n = B(1/3)$. For $x^2 + y^2 < 1/9$, we have $x^2 + y^2 < (1/3 - 1/2n)^2$ if n is large enough (and also $x^2 + y^2 < 1 + 1/n$ for all n). However, if $x^2 + y^2 \geq 1/3$, then (x, y) does not belong to any A_n with even n . Thus, $\limsup A_n \neq \liminf A_n$, and the sequence $\{A_n\}$ does not converge.

Infinite operations on events play a very important role in the development of the theory, especially in determining limiting probabilities.

The definitions below will prepare the ground for the considerations in the following chapters. In Chapter 2, we will introduce probability as a number assigned to an event. Formally, we will be considering numerical functions defined on events, that is, on subsets of the sample space $\mathcal{S}$. As long as $\mathcal{S}$ is finite or countably infinite, we can take the class of all subsets of $\mathcal{S}$ as the domain of definition of probability. In case of infinite but *not* countable $\mathcal{S}$ (e.g., where $\mathcal{S}$ is an interval, the real line, or a plane), it may not be possible to define probability on the class of *all* subsets of $\mathcal{S}$. Although the explanation lies beyond the scope of this book, we will show how the difficulties can be avoided by suitable restriction of the class of subsets of $\mathcal{S}$ that are taken as events. We begin with the concept of *closure* under some operation.

Definition 1.4.1 We say that the class $\mathcal{A}$ of subsets of $\mathcal{S}$ is *closed* under a given operation if the sets resulting from performing this operation on elements of $\mathcal{A}$ are also elements of $\mathcal{A}$. $\square$

Complementation A^c , finite union $A_1 \cup \dots \cup A_n$, infinite union $A_1 \cup A_2 \cup \dots$, limits of sequences $\lim A_n$, are few examples of such operations.

■ EXAMPLE 1.18

Let $\mathcal{S} = \{0, 1, 2, \dots\}$, and let $\mathcal{A}$ consist of all subsets of $\mathcal{S}$ that are finite. $\mathcal{A}$ is closed under finite unions and all intersections, finite or not. Indeed, if $A_1, \dots, A_n$ are finite sets, then $A = A_1 \cup \dots \cup A_n$ is also finite. Similarly, if $A_1, A_2, \dots$ are finite, then $\bigcap_i A_i \subset A_1$, and hence $\bigcap_i A_i$ is also finite. However, $\mathcal{A}$ is *not closed* under complementation: if A is finite ($A \in \mathcal{A}$), then A^c is not finite, and hence $A^c \notin \mathcal{A}$. On the other hand, if $\mathcal{A}$ is the class of all subsets of $\mathcal{S}$ that contain some fixed element, say 0, then $\mathcal{A}$ is closed under all intersections and unions, but it is not closed under complementation.

The following concepts have an important role in the theory of probability.

Definition 1.4.2 A nonempty class $\mathcal{A}$ of subsets of $\mathcal{S}$ that is closed under complementation and all finite operations (i.e., finite union, finite intersection) is called a *field*. If $\mathcal{A}$ is closed under complementation and all countable operations, it is called a *σ -field*. Finally, if $\mathcal{A}$ is closed under monotone passage to the limit,³ it is called a *monotone class*. $\square$

³In view of the fact proved earlier that all monotone sequences converge, this condition means that (a) if $A_1 \subset A_2 \subset \dots$ is an increasing sequence of sets in $\mathcal{A}$, then $\bigcup_i A_i \in \mathcal{A}$, and (b) if $A_1 \supset A_2 \supset \dots$ is a decreasing sequence of sets in $\mathcal{A}$, then $\bigcap_i A_i \in \mathcal{A}$.

Let us observe that Definition 1.4.2 can be formulated in a more efficient way. For $\mathcal{A}$ to be a field, it suffices to require that if $A, B \in \mathcal{A}$, then $A^c \in \mathcal{A}$ and $A \cap B \in \mathcal{A}$ (or $A^c \in \mathcal{A}$ and $A \cup B \in \mathcal{A}$). Any of these two conditions implies (by induction and De Morgan's laws) the closure of $\mathcal{A}$ under all finite operations. Consequently, for $\mathcal{A}$ to be a σ -field, it suffices to require that whenever $A_1, A_2, \dots \in \mathcal{A}$, then $A_i^c \in \mathcal{A}$ and $\bigcap_{i=1}^{\infty} A_i \in \mathcal{A}$ (or $A_i^c \in \mathcal{A}$ and $\bigcup_{i=1}^{\infty} A_i \in \mathcal{A}$); this follows again from De Morgan's laws.⁴

It is important to realize that closure under countable operations is stronger than closure under any finite operations. This means that there exist classes of sets that are fields but not σ -fields. This is illustrated by the following example:

■ EXAMPLE 1.19

Let $S = \{1, 2, 3, \dots\}$, and let $\mathcal{A}$ be the class of all subsets A of S such that either A or A^c is finite. Then $\mathcal{A}$ is a field but not a σ -field. First, if $A \in \mathcal{A}$, then $A^c \in \mathcal{A}$ because the definition of $\mathcal{A}$ is symmetric with respect to complementation. Next, if A and B are both in $\mathcal{A}$, so is their union. If A and B are both finite, then $A \cup B$ is finite and hence belongs to $\mathcal{A}$. On the other hand, if either A^c or B^c (or both) are finite, then $(A \cup B)^c = A^c \cap B^c$ is also finite because it is contained in A^c and also in B^c .

Thus, $\mathcal{A}$ is a field. However, $\mathcal{A}$ is not a σ -field. Let A_n be the set consisting only of the element n (i.e., $A_n = \{n\}$). Clearly, $A_n \in \mathcal{A}$. Take now $\bigcup_{n=1}^{\infty} A_{2n} = \{2, 4, 6, \dots\}$. This is a countable union of sets in $\mathcal{A}$ that is not in $\mathcal{A}$ since the set of all even numbers is not finite, nor does it have a finite complement.

Typically, it is easy to determine that a class of sets is a field, while direct verification that it is a σ -field can be difficult. On the other hand, it is sometimes easy to verify that a class of sets is a monotone class.

Theorem 1.4.2 *A σ -field is a monotone class. Conversely, a field that is a monotone class is a σ -field.*

Proof: To prove this theorem, assume first that $\mathcal{A}$ is a σ -field, and let $A_1, A_2, \dots$ be a monotone sequence of elements of $\mathcal{A}$. If $A_1 \subset A_2 \subset \dots$, then $\lim A_n = \bigcup_{n=1}^{\infty} A_n \in \mathcal{A}$, whereas if $A_1 \supset A_2 \supset \dots$, then $\lim A_n = \bigcap_{n=1}^{\infty} A_n \in \mathcal{A}$. So $\mathcal{A}$ is a monotone class. On the other hand, let $\mathcal{A}$ be a monotone class and a field, and let $A_1, A_2, \dots$ be an arbitrary sequence of elements of $\mathcal{A}$. Put $B_n = A_1 \cup \dots \cup A_n$. Then since $\mathcal{A}$ is a field, and also $B_1 \subset B_2 \subset \dots$, $B_n \in \mathcal{A}$ for every n . Further, since $\mathcal{A}$ is a monotone class, $\lim B_n \in \mathcal{A}$. However, $\lim B_n = \bigcup_{n=1}^{\infty} B_n = \bigcup_{n=1}^{\infty} A_n$, so $\mathcal{A}$ is a σ -field, as asserted. $\square$

The last in this series of concepts is that of the minimal field (or σ -field, or monotone class) containing a given set or collection of sets. We begin with some examples.

■ EXAMPLE 1.20

Let S be any set. On one extreme, the class consisting of two sets, $\emptyset$ and S , is closed under any operation so that $\mathcal{A} = \{\emptyset, S\}$ is a field, a σ -field, and a monotone class. On the other extreme, the class of *all* subsets of S is also closed under any operations, finite or not, and hence is a field, a σ -field, and a monotone class. These two classes of subsets of S form the smallest and the largest fields (σ -field, monotone class).

⁴For various relations among classes of sets defined through closure properties under operations, for example, see Chow and Teicher (1997) and Chung (2001).

For any event A , it is easy to check that the class $\mathcal{A}$, consisting of the four events $\{\emptyset, A, A^c, S\}$, is closed under any operations: unions, intersections, and complements of members of $\mathcal{A}$ are again members of $\mathcal{A}$. This class is an example of a field (σ -field, monotone class) that contains the events A and A^c , and it is the smallest such field (σ -field, monotone class).

On the other hand, the class $\mathcal{A}$, consisting of events $\{\emptyset, A, S\}$, is a monotone class, but neither a field nor σ -field. If A and B are two events, then the smallest field $\mathcal{A}$ containing A and B must contain also the sets A^c, B^c , the intersections $A \cap B$, $A \cap B^c$, $A^c \cap B$, $A^c \cap B^c$, as well as their unions $A \cup B$, $A \cup B^c$, $A^c \cup B$, and $A^c \cup B^c$. The closure property implies that unions such as $(A \cap B) \cup (A \cup B^c)$, must also belong to $\mathcal{A}$.

We are now ready to present the final step.

Theorem 1.4.3 *For any nonempty class $\mathcal{K}$ of subsets of S , there exists a unique smallest field (σ -field, monotone class) containing all sets in $\mathcal{K}$. It is called the field (σ -field, monotone class) generated by $\mathcal{K}$.*

Proof: We will prove the assertion for fields. Observe first that if $\mathcal{A}_1$ and $\mathcal{A}_2$ are fields, then their intersection $\mathcal{A}_1 \cap \mathcal{A}_2$ (i.e., the class of sets that belong to both $\mathcal{A}_1$ and $\mathcal{A}_2$) is also a field. For instance, if $A, B \in \mathcal{A}_i$ ($i = 1, 2$), then $A \cup B \in \mathcal{A}_i$ because each $\mathcal{A}_i$ is a field, and consequently $A \cup B \in \mathcal{A}_1 \cap \mathcal{A}_2$. A similar argument holds for intersections and complements.

Note that if $\mathcal{A}_1$ and $\mathcal{A}_2$ contain the class $\mathcal{K}$, then the intersection $\mathcal{A}_1 \cap \mathcal{A}_2$ also contains $\mathcal{K}$. The property extends to any intersection of fields containing $\mathcal{K}$ (not only the intersections of two such fields).

Now, let $\mathcal{C}$ be the intersection of *all* fields containing $\mathcal{K}$. We claim that $\mathcal{C}$ is the minimal unique field containing $\mathcal{K}$. We have to show that (1) $\mathcal{C}$ exists, (2) $\mathcal{C}$ is a field containing $\mathcal{K}$, (3) $\mathcal{C}$ is unique, and (4) $\mathcal{C}$ is minimal.

For property (1) it is enough to show that there exists at least one field containing $\mathcal{K}$. We may take here the class of all subsets of S : it is a field (as well as a σ -field and monotone class), and it contains all sets in $\mathcal{K}$. Property (2) follows from the fact that the intersection of fields containing $\mathcal{K}$ is a field containing $\mathcal{K}$. Property (3) (i.e., uniqueness of $\mathcal{C}$) follows from the fact that the result of the operation of intersection is unique.

Finally, suppose that there exists a field $\mathcal{C}'$ containing $\mathcal{K}$ such that $\mathcal{C}' \subset \mathcal{C}$. Then $\mathcal{C}'$ must appear as one of the factors in the intersection defining $\mathcal{C}$ so that $\mathcal{C} \subset \mathcal{C}'$. Consequently, $\mathcal{C}' = \mathcal{C}$. This completes the proof for the case of fields. The proofs for σ -fields and monotone classes are exactly the same, since an intersection of σ -fields (or monotone classes) containing $\mathcal{K}$ is again a σ -field (monotone class) containing $\mathcal{K}$. $\square$

One may find it disturbing that Theorem 1.4.3 asserts the existence and uniqueness of some objects without giving a clue as to how to practically find them. In fact, the nonconstructive character of the theorem, combined with its generality, is instead a great help. As we will see in Chapter 2, the natural objects of our interest (the domains of definition of probability) will be σ -fields of events. Beyond the trivial situations of finite or countably infinite sample spaces S , where one can always consider the maximal σ -field consisting of all subsets of S , one is forced to restrict consideration to classes of events that form σ -fields generated by some “simple” events. The events in these σ -fields are typically of a very rich structure, and one seldom has useful criteria for distinguishing events (elements of the σ -field in question) from “nonevents,” that is, subsets of S to which probabilities are not assigned. However, as shown by the two examples below, the smallest σ -field generated by some class is richer than the smallest field generated by the same class.

■ EXAMPLE 1.21

A point moves randomly on the plane, and its location is recorded at some time t . The outcome of this experiment is the pair (x, y) of coordinates of the observed location of the point (e.g., imagine here the location of a particle of dust in a liquid, tossed about by random hits from molecules of the medium, and performing Brownian motion; or imagine a location of a previously marked bird at the time of its capture in a bird migration study or the ages of both husband and wife at the time one of them dies).

In any study of this kind (regardless of its ultimate purpose), the “natural” sample space $\mathcal{S}$ is a plane or part of the plane, (the positive quadrant, etc.). The “simple” events here are of the form $a < x \leq b, c < y \leq d$, that is, rectangles with sides parallel to the axes. The reason for distinguishing these events as “simple” is that, as will be explained in later chapters, it is often easy to assign probabilities to these events. The reason for the particular configuration of strict and nonstrict inequalities (i.e., north and east side included, south and west side excluded) will also become apparent from the analysis below. To simplify the language, we will call such events Rectangles, and use a capital letter to signify the specific assumption about which sides are included and which are not. Naturally, we will allow for infinite Rectangles, such as $\{a < x \leq \infty, -\infty < y \leq b\}$.

It is easy to determine the field generated by all Rectangles: These are events that result from finite operations on Rectangles. Clearly, the complement of a Rectangle is a union of at most eight disjoint (infinite) Rectangles (see Figure 1.7), whereas the intersection of Rectangles is again a Rectangle (or is empty). Since unions are reduced to intersections of complements by De Morgan’s laws, every element of the smallest field containing all Rectangles is the union of a finite number of disjoint Rectangles. On the other hand, there exist events that do not belong to this field of events. As a simple example, one might be interested in the event that the point (x, y) lies within distance r from some fixed point (from the initial location of the particle, the point of release of the bird, etc.). This event is a circle on the plane, and hence a subset of $\mathcal{S}$, which is not decomposable into a finite number of Rectangles. On the other hand, a circle does belong to the σ -field spanned by Rectangles: it is representable as a countable union of Rectangles, or equivalently, as an infinite intersection of sets built up of Rectangles.

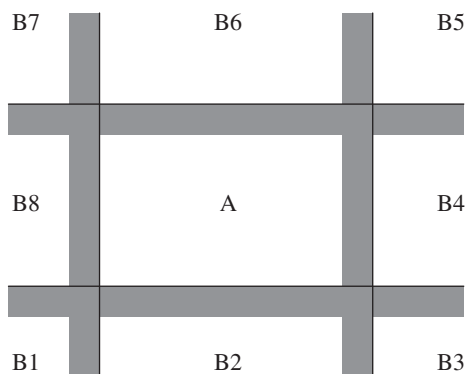


Figure 1.7 Complement of a Rectangle.

Similarly, in this example there are other events, which are not in the field generated by Rectangles and which could be considered, such as triangles, rectangles with sides not parallel to the axes, and ellipses.

■ **EXAMPLE 1.22**

Take an experiment consisting of tossing a coin infinitely many times. The “natural” sample space $\mathcal{S}$ is the space of all infinite sequences $x = (\xi_1, \xi_2, \dots)$, where $\xi_i = 0$ or 1 (or any other two distinct symbols representing heads and tails). The “simple” events here are of the form “heads on the n th toss,” that is, sets of all infinite sequences $x = (\xi_1, \xi_2, \dots)$ with the n th coordinate ξ_n satisfying $\xi_n = 0$. The events in the field generated by the simple events are of the form “heads on tosses $k_1, \dots, k_n$ and tails on tosses $r_1, \dots, r_m$,” with both m and n finite and the outcomes of all other tosses remaining unspecified.

An event that does not belong to this field, but does belong to the σ -field generated by the simple events, is the event that “as the number of tosses increases, the frequency of heads approaches a limit.” Clearly, to determine whether or not this event occurs, it does not suffice to know any finite number of coordinates ξ_n .

To generalize this example, replace the outcome of the coin tosses by the result of some experiment repeated infinitely many times. This way the coordinate ξ_n carries more information than it does for the outcome of n th coin toss. The “simple” events are now of the form of sets of sequences $x = (\xi_1, \xi_1, \dots)$ with $\xi_i \in A_i$ for $i = 1, \dots, n$, while the ξ_i ’s for $i > n$ are unconstrained. Here $A_1, \dots, A_n$ are events that occur at the first n times of observations. The “simple” events described above, of an obvious interest and importance both in applications and in building the theory, are called “cylinder” events. The smallest σ -field containing all cylinder events comprises all events that may be of interest, including those that are obtained through limits of sequences of cylinder events.

PROBLEMS

1.4.1 Let $B_1, B_2, \dots$ be a countable partition of $\mathcal{S}$; that is, $B_i \cap B_j = \emptyset$ for all $i \neq j$, and $\bigcup_i B_i = \mathcal{S}$. Let $A_n = B_n \cup B_{n+1} \cup \dots$. Find $\lim A_n$.

1.4.2 Assume that John will live forever. He plays a certain game each day. Let A_i be the event that he wins the game on the i th day. (i) Let B be the event that John will win every game starting on January 1, 2035. Label the following statements as true or false: (a) $B = \liminf A_n$. (b) $B \subset \liminf A_n$. (c) $B \supset \limsup A_n$. (d) $B = \limsup A_n$. (ii) Assume now that John starts playing on a Monday. Match the following events C_1 through C_9 with events D_1 through D_{11} :

C_1 = John loses infinitely many games.

C_2 = When John loses on a Thursday, he wins on the following Sunday.

C_3 = John never wins on three consecutive days.

C_4 = John wins every Wednesday.

C_5 = John wins on infinitely many Wednesdays.

C_6 = John wins on a Wednesday.

C_7 = John never wins on a weekend.

C_8 = John wins infinitely many games and loses infinitely many games.

C_9 = If John wins on some day, he never loses on the next day.

$D_1 = \bigcap_{j=0}^{\infty} [A_{7j+4} \cup A_{7(j+1)}],$

$D_2 = \bigcap_{j=0}^{\infty} A_{7j+3},$

$D_3 = \bigcap_{j=0}^{\infty} [A_{7j+6}^c \cap A_{7(j+1)}^c],$

$D_4 = \bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} A_k^c,$

$D_5 = \bigcap_{i=1}^{\infty} [A_i^c \cup A_{i+1}^c \cup A_{i+2}^c],$

$D_6 = \bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} A_{7k+3},$

$D_7 = \left[\bigcup_{j=0}^{\infty} [A_{7j+6} \cup A_{7(j+1)}] \right]^c,$

$D_8 = \bigcup_{j=0}^{\infty} A_{7j+3},$

$$D_9 = \bigcup_{n=0}^{\infty} \left\{ \left[\bigcap_{i=1}^n A_i^c \right] \cap \left[\bigcap_{k=n+1}^{\infty} A_k \right] \right\}, \quad D_{10} = \left\{ \bigcap_{i=0}^{\infty} [A_i \cap A_{i+1} \cap A_{i+2}] \right\}^c,$$

$$D_{11} = \left[\bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} A_k^c \right] \cap \left[\bigcap_{j=1}^{\infty} \bigcup_{m=j}^{\infty} A_m \right].$$

- 1.4.3** Let $A_1, \dots, A_n$ be distinct subsets of $\mathcal{S}$. **(i)** Find the maximum number of sets (including $\mathcal{S}$ and $\emptyset$) of the smallest field containing $A_1, \dots, A_n$. **(ii)** Find the maximum number of sets in this field if $A_{n-1} \subset A_n$. **(iii)** Answer (ii) if $A_1 \subset A_2 \subset \dots \subset A_n$. **(iv)** Answer (ii) if $A_1 = \dots = A_n = \emptyset$. **(v)** Answer (i)–(iv) for a σ -field.
- 1.4.4** For $0 < \alpha < 1$, let $I(\alpha) = \{x : 1 - \alpha < x < 1 + \alpha\}$. Consider a sequence $\alpha_1, \alpha_2, \dots$ of numbers satisfying $0 < \alpha_n < 1$ for all n , and let $A_n = I(\alpha_n)$. **(i)** Find $\limsup A_n$ and $\liminf A_n$. **(ii)** Find conditions, expressed in terms of α_n , under which $\lim A_n$ exists, and find this limit. **(iii)** Define $J(\alpha) = \{x : 1 - \alpha \leq x \leq 1 + \alpha\}$ and $B_n = J(\alpha_n)$. Answer questions (i) and (ii) for sequence $\{B_n\}$.
- 1.4.5** Let $\mathcal{S} = \{0, 1, 2, \dots\}$ be the set of all integers. For $A \subset \mathcal{S}$, let $f_n(A)$ be the number of elements in the intersection $A \cap \{0, 1, \dots, n\}$. Let $\mathcal{A}$ be the class of all sets A for which the limit

$$q(A) = \lim_{n \rightarrow \infty} \frac{f_n(A)}{n}$$

exists. Show that $\mathcal{A}$ is not a field. [*Hint:* Let $A_1 = \{1, 3, 5, \dots\}$ and $A_2 = \{\text{all odd integers between } 2^{2n} \text{ and } 2^{2n+1} \text{ and all even integers between } 2^{2n+1} \text{ and } 2^{2n+2} \text{ for } n = 0, 1, \dots\}$. Show that both A_1 and A_2 are in $\mathcal{A}$ but $A_1 \cap A_2 \notin \mathcal{A}$.]

- 1.4.6** Let $S = (-\infty, +\infty)$. Show that the class of all finite unions of intervals of the form $[a, b]$, (a, b) , $[a, b)$, and $(a, b]$, with possibly infinite a or b (intervals of the form $[a, \infty)$, etc.) forms a field.